

Input from computeCov2D (dilated)

$$\Sigma'' = \begin{pmatrix} \Sigma''_{00} & \Sigma'_{01} \\ \Sigma'_{01} & \Sigma''_{11} \end{pmatrix}$$

$$\det = \Sigma''_{00} \Sigma''_{11} - \Sigma'^2_{01}$$

$\det = 0 ?$

yes

Gaussian is culled  
(not represented)

no

$$\Sigma'^{-1} = \text{conic} = \frac{1}{\det} \begin{pmatrix} \Sigma''_{11} & -\Sigma'_{01} \\ -\Sigma'_{01} & \Sigma''_{00} \end{pmatrix}$$

stored as (conic.x, conic.y, conic.z) =

$$\left( \frac{\Sigma''_{11}}{\det}, \frac{-\Sigma'_{01}}{\det}, \frac{\Sigma''_{00}}{\det} \right)$$